

Manzil JEE 2026

Chemistry (Practice Sheet)

Some Basic Concept of Chemistry

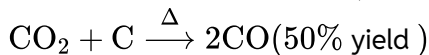
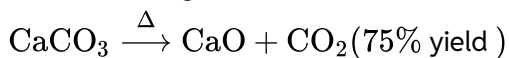
Time duration-150 min.

- Q1** A compound contains 11.99% N, 13.70% O, 9.25% B and 65.06% F. Its empirical formula (molar mass of B is 10.8 g mol^{-1})
- (A) NOBF_2
 (B) NOBF_4
 (C) N_2OF_2
 (D) $\text{NO}_2 \text{ F}_2$
- Q2** Woods metal contains 50.0% bismuth, 25.0% lead, 12.5% tin and 12.5% cadmium by weight. What is the mole fraction of tin ?
 (Atomic weights :
 $\text{Bi} = 209$, $\text{Pb} = 207$, $\text{Sn} = 119$, $\text{Cd} = 112$)
- (A) 0.202 (B) 0.158
 (C) 0.176 (D) 0.221
- Q3** Enzymes convert glucose ($M = 180.2$) to ethanol ($M = 46.1$) according to the equation
 $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$
 What is the maximum mass of ethanol that can be made from 15.5 kg of glucose?
- (A) 0.256 kg
 (B) 0.512 kg
 (C) 3.96 kg
 (D) 7.93 kg
- Q4** An element X has three isotopes ^{20}X , ^{21}X and ^{22}X . The percentage abundance of ^{20}X is 90% and average atomic mass of the element is 20.18. The percentage abundance of ^{21}X should be:
- (A) 2% (B) 8%
 (C) 10% (D) 0%
- Q5** An aqueous solution of $\text{C}_6\text{H}_{12}\text{O}_6$ is 10% w/v. The volume in which 1 mole of it is dissolved will be:
- (A) 0.8 L
 (B) 16 L
 (C) 23 L
 (D) 1.8 L
- Q6** The mass of 3.2×10^5 atoms of an element is $8.0 \times 10^{-18} \text{ gm}$. The atomic mass of the element in (g) is about
- (A) 2.5×10^{-22}
 (B) 15
 (C) 8.0×10^{-18}
 (D) 30
- Q7** Generally commercial hydrochloric acid is prepared by heating NaCl with concentrated H_2SO_4 . How much H_2SO_4 solution containing 93.0% H_2SO_4 by mass is required for the production of 1000 kg of concentrated hydrochloric acid containing 43% HCl by weight?
- (A) 590.0 kg solution of H_2SO_4
 (B) 310.3 kg solution of H_2SO_4
 (C) 620.7 kg solution of H_2SO_4
 (D) 708.2 kg solution of H_2SO_4
- Q8** Haemoglobin contains 0.33% of iron by weight. The molecular weight of haemoglobin is approximately 67200. The number of iron atoms (at. wt. of Fe = 56) present in one molecule of haemoglobin is
- (A) 6 (B) 1
 (C) 2 (D) 4
- Q9** The simplest formula of a compound containing 50% of element X (atomic mass 10) and 50% of element Y (atomic mass 20) is:
- (A) XY



- (B) X_2Y
 (C) XY_3
 (D) X_2Y_3

Q10 In the following set of reactions



CO formed at STP from 10 g of $CaCO_3$ is

- (A) 2.24 L
 (B) 11.2 L
 (C) 1.12 L
 (D) 1.68 L

Q11 Equal masses of oxygen, hydrogen and methane are taken in identical conditions. What is the ratio of the volumes of the gases under identical conditions?

- (A) 16 : 1 : 8 (B) 1 : 16 : 2
 (C) 1 : 16 : 8 (D) 2 : 16 : 1

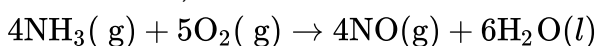
Q12 The density of a 56.0% by mass aqueous solution of 1-propanol ($CH_3CH_2CH_2OH$) is 0.8975 g/cm^3 . What is the mole fraction of the 1-propanol?

- (A) 0.292 (B) 0.227
 (C) 0.241 (D) 0.276

Q13 A compound contains equal masses of the elements A, B and C. If the atomic masses of A, B and C are 20, 40 and 60 respectively, then the empirical formula of the compound is :

- (A) $A_3 B_2 C$
 (B) AB_2C_3
 (C) ABC
 (D) $A_6 B_3 C_2$

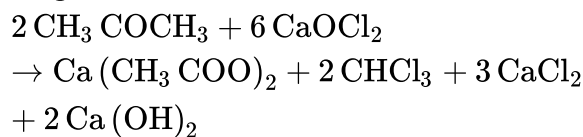
Q14 In the reaction,



When 1 mol of ammonia and 1 mol of O_2 are made to react to completion then:

- (A) 1.0 mol of H_2O is produced
 (B) 1.0 mol of NO will be produced
 (C) all the oxygen will be consumed
 (D) all the ammonia will be consumed

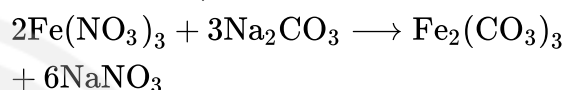
Q15 If the yield of chloroform obtainable from acetone and bleaching powder is 75%, then what mass of acetone is required for producing 30 g of chloroform?



[Molar mass of $CHCl_3$ = 119.5; Molar mass of CH_3COCH_3 = 58]

- (A) 40 g
 (B) 19.4 g
 (C) 10.92 g
 (D) 14.56 g

Q16 For the reaction,



Initially if 2.5 mole of $Fe(NO_3)_3$ and 3.6 mole of Na_2CO_3 is taken. If 6.3 mole of $NaNO_3$ is obtained then % yield of given reaction is

- (A) 50 (B) 84
 (C) 87.5 (D) 100

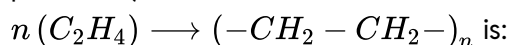
Q17 A solution of ethanol in water is 10% by volume. If the solution and pure ethanol have densities of 0.9866 g/cc and 0.785 g/cc respectively. The percent by weight is nearly?

- (A) 7.95% (B) 17%
 (C) 9.86% (D) 16.2%

Q18 Rearrange the following (I to IV) in the order of increasing masses:

- (I) 0.5 moles of CO_2
 (II) 0.5gm atom of oxygen
 (III) 3.011×10^{23} molecules of O_2
 (IV) 5.6 litre of CO_2 at 1 atm and 273 K
 (A) II < IV < III < I
 (B) II < I < IV < III
 (C) IV < II < III < I
 (D) II < III < IV < I

Q19 When 100 g of ethylene polymerises entirely to polyethene, the weight of polyethene formed as per the equation



- (A) $(n/2)g$ (B) 100g
(C) $(100/n)g$ (D) 100ng

Q20 A metal chloride contains 55.0% of chlorine by weight. 100 mL vapours of the metal chloride at STP weigh 0.57 g. The molecular formula of the metal chloride is (Given: Atomic mass of chlorine is 35.5 u)

- (A) MCl_2 (B) MCl_4
(C) MCl_3 (D) MCl

Q21 25.4 g of iodine and 14.2 g of chlorine are made to react completely to yield a mixture of ICl and ICl_3 . Calculate the number of moles of ICl and ICl_3 formed.

- (A) 0.1 mole, 0.1 mole
(B) 0.1 mole, 0.2 mole
(C) 0.5 mole, 0.5 mole
(D) 0.2 mole, 0.2 mole

Q22 Among the following pairs of compounds, the one that illustrates the law of multiple proportions is

- (A) NH_3 and NCl_3
(B) H_2S and SO_2
(C) CuO and Cu_2O
(D) CS_2 and $FeSO_4$

Q23 Average atomic mass of magnesium is 24.31 amu. This magnesium is composed of 79 mole % of ^{24}Mg and remaining 21 mole % of ^{25}Mg and ^{26}Mg . Calculate mole % of ^{26}Mg .

- (A) 10 (B) 11
(C) 15 (D) 16

Q24 For the given reaction $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

- (1) If 64 g of O_2 is used, then 44 g of CO_2 is formed.
(2) 8 g of CH_4 reacts to form 36 g of product.
(3) 22 g of CO_2 is formed from 3.011×10^{23} molecules of CH_4 .
(4) At STP, if 22.4 litres of $O_2(g)$ is used, then 11.2 litres of CO_2 is formed.

Which of the above statements are correct ?

- (A) 2, 3, 4 (B) 1, 2, 3

- (C) 1, 3, 4 (D) 1, 3 only

Q25 A quantity of 10 g of a hydrocarbon exactly requires 40 g oxygen for complete combustion. The products formed are CO_2 and water. When CO_2 gas formed is absorbed completely in lime water, the mass of solution increases by 27.5 g. What is the mass of water formed simultaneously in the combustion?

- (A) 22.5 g
(B) 27.5 g
(C) 50 g
(D) 10 g

Q26 Identify the option containing maximum number of atoms.

- (A) 18 mg of glucose
(B) 2 mg of hydrogen gas
(C) 10 mg of H_2O
(D) 7.8 mg of benzene

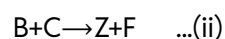
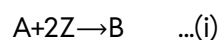
Q27 How a sample of calcium carbonate ($CaCO_3$) has the following percentage composition :

Ca = 40%, C = 12%, O = 48%

If the law of constant proportions is true, then the weight of calcium in 4g of a sample of calcium carbonate obtained from another source will be (Gram atomic mass of Ca = 40 g, O = 16 g and C = 12 g)

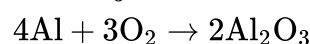
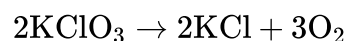
- (A) 0.016 g (B) 0.16 g
(C) 1.6 g (D) 16 g

Q28 5 moles of A, 6 moles of Z are mixed with sufficient amount of C to finally produce F. Then find the maximum moles of F which can be produced. Assuming that the product formed can also be reused. Reactions are



- (A) 3 moles (B) 4.5 moles
(C) 5 moles (D) 6 moles

Q29 Based on the following consecutive reactions

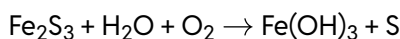


2 moles of KClO_3 of 50% purity on strong heating and then with excess of Al , Al_2O_3 formed is

- (A) 2 mol
(B) 1 mol
(C) 3 mol
(D) 1.5 mol

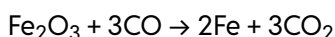
Q30 If $\text{P}_2\text{O}_7^{4-}$ anion contains a total of 180 moles of electrons, find the number of moles of $\text{P}_2\text{O}_7^{4-}$ present.

Q31 Mixture of 10 moles of Fe_2S_3 , 20 moles of H_2O and 30 moles of O_2 react with 30% yield of given reaction:



Calculate moles of $\text{Fe}(\text{OH})_3$ that can be produced in above reaction.

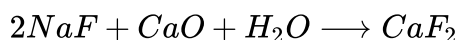
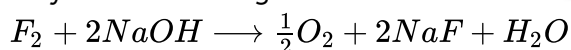
Q32 Moles of iron which can be made from Fe_2O_3 using 294 grams of CO in the reaction:



(Given that the molar mass of $\text{Fe}_2\text{O}_3 = 160$, $\text{CO} = 28$)

Q33 An organic compound gives 0.220 g of CO_2 and 0.126 g of H_2O on complete combustion. If the % of carbon is 24, then the % hydrogen is $\times 10^{-1}$. (Nearest integer)

Q34 A fluorine disposal plant was constructed to carryout the following reactions:



Over a period of operation, 1900 kg of fluorine was fed into a plant and 10,000 kg of lime was required. What was the percentage utilisation of lime? [Lime : CaO]

Q35 $\text{Cl}_2 + \text{KOH} \xrightarrow{60\%} \text{KCl} + \text{KClO} + \text{H}_2\text{O}$



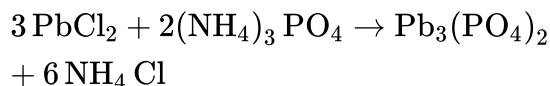
112 L Cl_2 gas at STP is passed in 10 L KOH solution, containing 1 mole of potassium

hydroxide per liter.

Calculate the total moles of KCl produced, rounding it off to the nearest whole number (Yield of chemical reactions are written above the arrow ($\rightarrow$) of respective reaction)

Q36 Aspartame, an artificial sweetener contains 9.52 weight % of nitrogen. There are two nitrogen atoms per molecule. What is the molecular weight of aspartame? (Nearest Integer) (Given: Atomic mass of N = 14)

Q37 Consider the following reaction



If 72 mmol of PbCl_2 is mixed with 50 mmol of $(\text{NH}_4)_3\text{PO}_4$,

then the amount of $\text{Pb}_3(\text{PO}_4)_2$ formed is ____ mmol (nearest integer)

Q38 10 g of hydrogen and 64 g of oxygen were filled in a steel vessel and exploded. Amount of water (in g) produced in this reaction will be:

Q39 The ratio of the mass percentages of 'C & H' and 'C & O' of a saturated acyclic organic compound 'X' are 4 : 1 and 3 : 4 respectively. Then, the moles of oxygen gas required for complete combustion of two moles of organic compound 'X' is (Given : vapor density of the compound is 31)

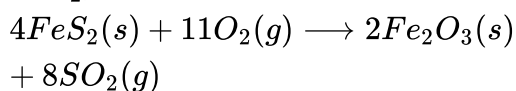
Q40 A sample contains 3.01×10^{20} molecules of SF_n and its mass is 54mg. What is the value of n ?

Q41 Two elements A and B forms 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.

Q42 $\text{SO}_2 + 2\text{H}_2\text{S} \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
2 mol of H_2S and 11.2 L SO_2 at STP react to form x mol of Sulphur. The value of 10x is ____



- Q43** Pure iron pyrite, FeS_2 , is burnt with 50% excess air than required for complete oxidation of FeS_2 , in a closed vessel.



Air contains 20% O_2 and 80% N_2 , by volume.

The mole percent of N_2 gas in the gaseous mixture, after complete reaction, is.

- Q44** A solution has 80% w/w NaOH with density 2 g L^{-1} then molality of solution is _____ mol kg^{-1}

- Q45** An alkaloid contains 17.28% of nitrogen and it's molecular mass is 162. The number of nitrogen atoms present in one molecule of alkaloid is:

- Q46** Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)

(Given: Atomic mass of Mg is 24 g mol^{-1} ; $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)

- Q47** A herbal paste was partially dried. The dried paste contained 70% active compound and 5% water. The originally, the paste contained 25% water. What is the percentage of active compound in the original paste. (Nearest Integer).

- Q48** Sample of an element "X" consist of its three isotopes A_1 , A_2 & A_3 and population of A_2 is three times the population of A_3 . If the average atomic mass of the sample is 1.25 u then determine percentage population of A_1 . (Atomic masses of isotopes A_1 , A_2 & A_3 are 1 u, 2 u and 3 u respectively.)

- Q49** The answer is a non-negative integer.
5 g of K_2SO_4 were dissolved in 250 mL of solution. The volume of this solution that should be used so that 1.2 g of $BaSO_4$ may be precipitated from $BaCl_2$ is

_____ (Given that molar mass of $K_2SO_4 = 174 \text{ g mol}^{-1}$ and $BaSO_4 = 233 \text{ g mol}^{-1}$.)

- Q50** A solution contain substances A and B in H_2O (solvent). The mole fraction of 'A' is 0.05 and molarity of 'B' is 7 M. The solution has density 1.14 gm/ml. Calculate "molarity of A". [Molecular weight of A = 10 g/mol; molecular weight of B = 30 g/mol, $H_2O = 18 \text{ g/mol}$]



Answer Key

Q1 (B)
Q2 (C)
Q3 (D)
Q4 (A)
Q5 (D)
Q6 (B)
Q7 (C)
Q8 (D)
Q9 (B)
Q10 (D)
Q11 (B)
Q12 (D)
Q13 (D)
Q14 (C)
Q15 (B)
Q16 (C)
Q17 (A)
Q18 (A)
Q19 (B)
Q20 (A)
Q21 (A)
Q22 (C)
Q23 (A)
Q24 (C)
Q25 (A)

Q26 (A)
Q27 (C)
Q28 (C)
Q29 (B)
Q30 2
Q31 4
Q32 7
Q33 56
Q34 28
Q35 4
Q36 294
Q37 24
Q38 72
Q39 5
Q40 4
Q41 2
Q42 15
Q43 83
Q44 100
Q45 2
Q46 24
Q47 55
Q48 80
Q49 45
Q50 3



Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Elements	Percentage	At. mass	Relative no. of atoms	Simplest ratio
N	11.99	14	$\frac{11.99}{14}$	1
O	13.70	16	$\frac{13.70}{16}$	1
B	9.25	10.8	$\frac{9.25}{10.8}$	1
F	65.06	19	$\frac{65.06}{19}$	4

Empirical formula = NOBF_4

Video Solution:



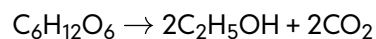
Q2 Text Solution:

$$X_{Sn} = \frac{12.5/119}{\frac{50}{209} + \frac{25}{207} + \frac{12.5}{119} + \frac{12.5}{112}} = 0.176$$

Video Solution:



Q3 Text Solution:



$$W_{\text{C}_6\text{H}_{12}\text{O}_6} = 15.5 \times 1000 = 15500 \text{ g}$$

$$M_{\text{C}_6\text{H}_{12}\text{O}_6} = 180.2 \text{ g/mol}$$

$$n_{\text{C}_6\text{H}_{12}\text{O}_6} = \frac{W_{\text{C}_6\text{H}_{12}\text{O}_6}}{M_{\text{C}_6\text{H}_{12}\text{O}_6}} = \frac{15500}{180.2} = 86.0155$$

According to stoichiometry of reaction, 1 mole of $\text{C}_6\text{H}_{12}\text{O}_6$ will form 2 mole of $\text{C}_2\text{H}_5\text{OH}$.

So, 86.0155 mole of $\text{C}_6\text{H}_{12}\text{O}_6$ will form 172.031 mole of $\text{C}_2\text{H}_5\text{OH}$.

$$\begin{aligned} \text{Mass of } \text{C}_2\text{H}_5\text{OH} &= 172.031 \times 46.1 \\ &= 7.93 \text{ kg} \end{aligned}$$

Video Solution:



Q4 Text Solution:

Percentage of abundance of $X^{20} = 90\%$

Let, percentage abundance of $X^{21} = y\%$

So, percentage abundance of $X^{22} = (10 - y)\%$

Average atomic mass =

$$20.18 = \frac{\Sigma \% \text{ abundance} \times \text{isotopic mass}}{100} = \frac{90 \times 20 + y \times 21 + (10 - y) \times 22}{100}$$

$$2018 = 1800 + 21y + 220 - 22y$$

$$y = 2020 - 2018 = 2$$

Video Solution:



Q5 Text Solution:

10% w/v of aqueous solution of $\text{C}_6\text{H}_{12}\text{O}_6$ means 10 g of $\text{C}_6\text{H}_{12}\text{O}_6$ in 100 mL solution.

So, 10 g of $\text{C}_6\text{H}_{12}\text{O}_6$ dissolved = 100 mL solution

1 g of $\text{C}_6\text{H}_{12}\text{O}_6$ dissolved = 10 mL solution.

180 g of $\text{C}_6\text{H}_{12}\text{O}_6$ dissolved = 1800 mL solution = 1.8 L

Video Solution:



Q6 Text Solution:

$$3.2 \times 10^5 \text{ atoms has mass} = 8 \times 10^{-18} \text{ g}$$



$$1 \text{ atom has mass} = \frac{8 \times 10^{-18}}{3.2 \times 10^5} \text{ g}$$

$$6.023 \times 10^{23} \text{ atoms has mass} =$$

$$\frac{8 \times 10^{-18} \times 6.023 \times 10^{23}}{3.2 \times 10^5} \text{ g}$$

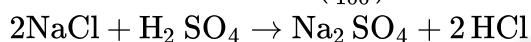
$$= 15 \text{ g}$$

Video Solution:



Q7 Text Solution:

$$\text{Mass of HCl} = 1000 \times \left(\frac{43}{100}\right) = 430 \text{ kg.}$$



$$\frac{\text{Mole of HCl}}{2} = \frac{\text{Mole of H}_2\text{SO}_4}{1}$$

$$\frac{430 \times 10^3}{36.5 \times 2} = \text{mole of H}_2\text{SO}_4$$

$$\text{Mass of H}_2\text{SO}_4 = \frac{98 \times 430 \times 10^3}{36.5 \times 2}$$

$$= 577.26 \times 10^3 \text{ g}$$

$$\text{Mass of 93\% H}_2\text{SO}_4 = 577.26 \times \frac{100}{93}$$

$$= 620.71 \text{ kg.}$$

Video Solution:



Q8 Text Solution:

$$\text{Mass of 1 molecule of Hb} = 67200$$

$$\text{Mass of iron in 1 molecule of Hb} = \frac{0.33}{100} \times 67200$$

$$\text{No. of iron atom in 1 molecule of Hb} =$$

$$\frac{0.33 \times 672}{56} = 3.96 \simeq 4$$

Video Solution:



Q9 Text Solution:

(B)

Let compound $\rightarrow 100 \text{ g}$

$$50 \text{ g} \rightarrow X$$

$$50 \text{ g} \rightarrow Y$$

$$n_X = \frac{50}{10} = 5$$

$$n_Y = \frac{50}{20} = 2.5$$

$$X : Y$$

$$5 : 2.5$$

$$2 : 1$$

So, simplest formula of compound = X_2Y

[NCERT Edition (2025-26) 11th Page No. 19]

Video Solution:



Q10 Text Solution:

$$10 \text{ g CaCO}_3 = 0.10 \text{ mol CO}_2 \text{ at } 75\% = 0.075$$

$$\text{mol; then } \text{CO}_2 + \text{C} \rightarrow 2\text{CO} \text{ at } 50\% \Rightarrow \text{CO} = 0.075$$

$$\text{mol} \Rightarrow \text{volume} = 0.075 \times 22.4 = 1.68 \text{ L.}$$

[NCERT Edition (2025-26) 11th Page No. 20]

Video Solution:



Q11 Text Solution:

Avogadro's law: Under identical conditions, equal volume of gases contains equal no. of moles.

$$m_{\text{O}_2} = m_{\text{H}_2} = m_{\text{CH}_4} = x \text{ g}$$

$$\text{mass ratio} = \text{O}_2 : \text{H}_2 : \text{CH}_4$$

$$\begin{array}{ccccccc} & 1 & : & 1 & : & 1 & \\ \text{mole ratio} = & 32 \times \frac{1}{32} & : & 32 \times \frac{1}{2} & : & 32 & \\ & \times \frac{1}{16} & & & & & \end{array}$$

$$\begin{array}{ccccccc} & 1 & : & 16 & : & 2 & \\ \text{Volume ratio} = & 1 & : & 16 & : & 2 & \end{array}$$

Video Solution:



**Q12 Text Solution:****Given:**

- **Mass % of 1-propanol** = 56.0%
→ That means **56 g 1-propanol + 44 g water** in 100 g solution
- **Density** = 0.8975 g/cm³
- We need to find **mole fraction of 1-propanol**.

Step 1: Moles of 1-propanolMolar mass of 1-propanol (CH₃CH₂CH₂OH):

$$= 3 \times 12.01 + 8 \times 1.008 + 16.00 \\ = 60.1 \text{ g/mol}$$

$$\text{Moles of 1-propanol} = \frac{56.0}{60.1} \approx 0.932 \text{ mol}$$

Step 2: Moles of water

Molar mass of water = 18.02 g/mol

$$\text{Moles of water} = \frac{44.0}{18.02} \approx 2.442 \text{ mol}$$

Step 3: Mole fraction of 1-propanol

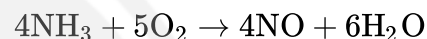
$$\chi_{\text{1-propanol}} = \frac{\text{moles of 1-propanol}}{\text{total moles}} \\ = \frac{0.932}{0.932 + 2.442} = \frac{0.932}{3.374} \approx 0.276$$

Video Solution:**Q13 Text Solution:**

If the compound has equal masses of A, B, and C, let each be 1g.

- A (atomic mass = 20) $\Rightarrow$ moles = $1/20 = 0.05$
- B (atomic mass = 40) $\Rightarrow$ moles = $1/40 = 0.025$
- C (atomic mass = 60) $\Rightarrow$ moles = $1/60 \approx 0.0167$

Divide by the smallest (0.0167), giving A:B:C = 3:1.5:1. Multiplying by 2 yields the integer ratio 6:3:2. Hence, **A₆B₃C₂**.

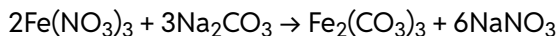
Video Solution:**Q14 Text Solution:**4 moles of NH₃ require 5 moles of O₂1 mol of NH₃ will require = $\frac{5}{4}$ mol of O₂But given O₂ $\Rightarrow$ 1 mol $\Rightarrow$ O₂ is limiting Reagent5 mole O₂ $\rightarrow$ 6 mol H₂O1 mole O₂ $\rightarrow$ $\frac{6}{5}$ mol H₂O1 mole O₂ $\rightarrow$ $\frac{4}{5}$ mol NO**Video Solution:****Q15 Text Solution:**Molar mass of CHCl₃ = 119.5Molar mass of CH₃COCH₃ = 5858 gm of CH₃COCH₃ produced 119.5 gmYield is 75% = $\frac{119.5 \times 75}{100} = 89.625$ gm

So, 89.625 produced by 58 gm

30 gm of chloroform is formed by

$$= \frac{58}{89.625} \times 30 = 19.4 \text{ gm}$$

Video Solution:

**Q16 Text Solution:**

According to stoichiometry of reaction, 2 mole of $\text{Fe}(\text{NO}_3)_3$ will react with 3 mole of Na_2CO_3 .

So, 2.5 mole of $\text{Fe}(\text{NO}_3)_3$ will react with 3.75 mole of Na_2CO_3 .

So, Na_2CO_3 will be limiting reagent because we have only 3.6 moles of Na_2CO_3

Now, 3 mole of Na_2CO_3 form 6 mole of NaNO_3

So, 3.6 mole of Na_2CO_3 will form 7.2 mole of NaNO_3 .

$$\begin{aligned}\% \text{ yield} &= \frac{\text{Actual yield}}{\text{Theoretical yield}} \times 100 \\ &= \frac{6.3}{7.2} \times 100 \\ &= 87.5\%\end{aligned}$$

Video Solution:**Q17 Text Solution:**

Given:

- Volume % of ethanol = 10%
- Density of solution = 0.9866 g/cc
- Density of pure ethanol = 0.785 g/cc

We need to calculate the **percent by weight** of ethanol in the solution.

Step 1: Assume 100 mL of solution

Since the solution is 10% by volume ethanol, in **100 mL of solution:**

- Volume of ethanol = 10 mL
- Volume of water = 90 mL

Step 2: Mass of ethanol

We know the density of pure ethanol = 0.785 g/cc (g/mL), so the mass of ethanol is:

$$\begin{aligned}\text{Mass of ethanol} &= \text{Volume} \times \text{Density} \\ &= 10 \text{ mL} \times 0.785 \text{ g/mL} = 7.85 \text{ g}\end{aligned}$$

Step 3: Mass of solution

The density of the solution is 0.9866 g/cc (g/mL), so the mass of the **100 mL solution** is:

$$\begin{aligned}\text{Mass of solution} &= 100 \text{ mL} \times 0.9866 \text{ g/mL} \\ &= 98.66 \text{ g}\end{aligned}$$

Step 4: Mass of water

The mass of water = Total mass of solution - Mass of ethanol:

$$\text{Mass of water} = 98.66 \text{ g} - 7.85 \text{ g} = 90.81 \text{ g}$$

Step 5: Percent by weight of ethanol

$$\begin{aligned}\text{Percent by weight of ethanol} &= \frac{\text{Mass of ethanol}}{\text{Mass of solution}} \times 100 = \frac{7.85}{98.66} \times 100 \\ &\approx 7.95\%\end{aligned}$$

Video Solution:**Q18 Text Solution:**

$$\begin{aligned}\text{(I) } w_{\text{CO}_2} &= n_{\text{CO}_2} \times M_{\text{CO}_2} \\ &= 0.5 \times 44 \\ &= 22 \text{ g}\end{aligned}$$

(II) 0.5 g atom = 0.5 mol of O- atom

$$W_{\text{O}} = 0.5 \times 16 = 8 \text{ g}$$

$$\begin{aligned}\text{(III) } n_{\text{O}_2} &= \frac{3.011 \times 10^{23} \text{ molecules}}{6.022 \times 10^{23} \text{ molecules mol}^{-1}} = \\ &= \frac{1}{2} \text{ mol}\end{aligned}$$

$$W_{\text{O}_2} = \frac{1}{2} \times 32 = 16 \text{ g}$$

(IV) $V_m = 22.4 \text{ L}$

$$n_{\text{CO}_2} = \frac{5.6}{22.4} = \frac{1}{4} \text{ mol}$$

$$w_{\text{CO}_2} = \frac{1}{4} \times 44 = 11 \text{ g}$$

$$\text{II} < \text{IV} < \text{III} < \text{I}$$



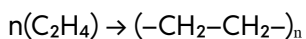
Video Solution:



Q19 Text Solution:

Given: 100 g of ethylene (C_2H_4) polymerises completely.

In polymerisation:



$\therefore$ No by-products formed, mass remains conserved.

So, if 100 g ethylene is used and 100% polymerised, then polyethylene formed = 100 g
Hence, correct option: B.

Video Solution:



Q20 Text Solution:

Given: 55% Cl by mass; 100 mL vapour weighs 0.57 g at STP; Cl = 35.5

Step 1:

$$\text{Molar mass} = \frac{0.57}{100/22400} = 127.7 \text{ g/mol}$$

Step 2:

Let formula = MCl_n

$$\frac{35.5n}{127.7} \times 100 = 55 \Rightarrow n \approx 2$$

Molecular formula: MCl_2

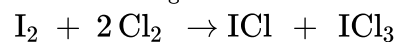
Video Solution:



Q21 Text Solution:

$$n_{I_2} = \frac{25.4 \text{ g}}{254 \text{ g mol}^{-1}} = 0.1 \text{ mol}$$

$$n_{Cl_2} = \frac{14.2 \text{ g}}{71 \text{ g mol}^{-1}} = 0.2 \text{ mol}$$



$$1 \text{ mol } I_2 = 2 \text{ mol } Cl_2$$

$$0.1 \text{ mol } I_2 = 0.2 \text{ mol } Cl_2$$

1 mol I_2 on reaction with 2 mol Cl_2 gives $ICl = 1 \text{ mol}$

0.1 mol $I_2 + 0.2 \text{ mol } Cl_2$ will form $ICl = 0.1 \text{ mol}$

similarly,

0.1 mol $I_2 + 0.2 \text{ mol } Cl_2$ will form $ICl_3 = 0.1 \text{ mol}$

Video Solution:



Q22 Text Solution:

CuO and Cu_2O illustrates law of multiple proportion.

Video Solution:



Q23 Text Solution:

Let the mole % of ^{26}Mg be x .

Given: Mole % of $^{24}Mg = 79\%$

Mole % of $^{25}Mg = 21\% - x$

Mole % of $^{26}Mg = x$

Average atomic mass:

$$79 \times 24 + (21 - x) \times 25 + x \times 26 = 24.31 \times 100$$

$$1896 + 525 - 25x + 26x = 2431$$

$$2421 + x = 2431$$



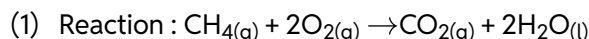
$$x = 10$$

Mole % of $^{26}\text{Mg} = 10$.

Video Solution:

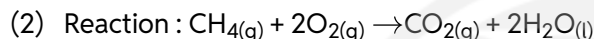


Q24 Text Solution:



64 g O_2 produces 44 gram CO_2

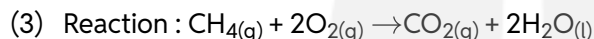
$\therefore$ (1) is correct (1)



$\therefore$ 16 gram CH_4 react to form $(44 + 36) = 80$ gram product

$$\frac{80 \times 8}{16} \text{ gram product} = 40 \text{ gram product}$$

$\therefore$ (2) is not correct



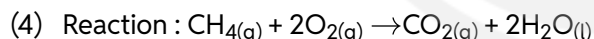
$\therefore$ 1 mole CH_4 form 1 mole CO_2 gas

$\therefore 6.022 \times 10^{23}$ molecules of CH_4 form 44 g CO_2 gas and

$\therefore 3.011 \times 10^{23}$ molecules of CH_4 form

$$= \frac{3.011 \times 10^{23} \times 44}{6.022 \times 10^{23}} = 22 \text{ g of } \text{CO}_2$$

(3) is correct



$\therefore$ At STP used 2 (22.4L) O_2 form 22.4 L CO_2

$\therefore$ At STP used 22.4 L O_2 form

$$= \frac{22.4\text{L} \times 22.4\text{L}}{2 \times 22.4\text{L}} \text{CO}_2 = 11.2 \text{ of } \text{CO}_2$$

(4) is correct

Thus (1), (3) and (4) are correct statements

Video Solution:



Q25 Text Solution:

Given:

- Mass of hydrocarbon = 10 g
- Mass of O_2 used = 40 g
- Mass of CO_2 formed = 27.5 g
- Products: CO_2 and H_2O
- Find mass of water formed.

Step-by-step (short):

Step 1: Total mass before combustion

$$\text{Initial mass} = 10 \text{ g (hydrocarbon)} + 40 \text{ g (oxygen)} = 50 \text{ g}$$

Step 2: Mass after combustion

$$\text{Total products} = \text{CO}_2 + \text{H}_2\text{O}$$

Mass of $\text{CO}_2 = 27.5 \text{ g}$

Let mass of $\text{H}_2\text{O} = x$

Apply conservation of mass:

$$27.5 + x = 50 \Rightarrow x = 22.5 \text{ g}$$

Final Answer: 22.5 g

Video Solution:



Q26 Text Solution:

$$W_{\text{glucose}} = 18 \times 10^{-3}, \text{Glucose} \Rightarrow \text{C}_6\text{H}_{12}\text{O}_6$$

$$M_{\text{glucose}} = 180 \text{ g/mol}$$

$$n_{\text{glucose}} = \frac{W_{\text{glucose}}}{M_{\text{glucose}}} = \frac{18 \times 10^{-3}}{180} = 10^{-4} \text{ mole}$$

$$\text{No. of atoms} = 24 \times 10^{-4} N_A \text{ atoms}$$

Video Solution:



Q27 Text Solution:

- A compound: Calcium carbonate (CaCO_3)
- Its percentage composition:
 - Calcium (Ca) = 40%
 - Carbon (C) = 12%
 - Oxygen (O) = 48%
- Mass of a new sample = 4 g

Step-by-step:

According to the Law of Constant Proportions, every sample of CaCO_3 will have 40% calcium, no matter the source.

So, in a 4 g sample:

$$\text{Mass of calcium} = \frac{40}{100} \times 4 = 1.6 \text{ g}$$

Video Solution:**Q28 Text Solution:**

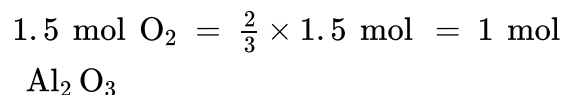
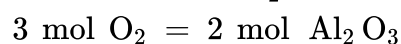
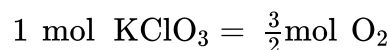
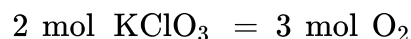
Net of (i)+(ii): $\text{A} + \text{C} + \text{Z} \rightarrow \text{F}$ (1 Z consumed per A). With 5 A and 6 Z, max

$\text{F} = \min(5, 6) = 5 \text{ mol}$ (Z formed is reused).

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Video Solution:**Q29 Text Solution:**

2 moles of KClO_3 contains 50% purity means 1 mol KClO_3 and 1 mol impurity

**Video Solution:****Q30 Text Solution:**

$\text{P}_2\text{O}_7^{4-}$ is known as pyrophosphate anion.

Total electrons in one $\text{P}_2\text{O}_7^{4-}$ ion

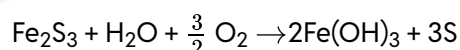
$= (2 \times 15 + 7 \times 8) + 4$ (due to 4 negative charge)

$= 90$ electrons.

90 mole of electrons will be present in $180 / 90$

mole $\text{P}_2\text{O}_7^{4-}$ ion;

$= 2/1 = 2$ moles of $\text{P}_2\text{O}_7^{4-}$ ion.

Video Solution:**Q31 Text Solution:**

According to stoichiometry of reaction,

1 mole of Fe_2S_3 reacts with 3 mole of H_2O and $\frac{3}{2}$ mole of O_2 .

So, 10 mole of Fe_2S_3 reacts with 30 mole of H_2O and 15 mole of O_2 .

So, H_2O is limiting reagent.

So, 3 mole of H_2O will form 2 mole of $\text{Fe}(\text{OH})_3$.

So, 20 mole of H_2O will form $\frac{40}{3}$ mole of $\text{Fe}(\text{OH})_3$.

$$\% \text{ yield} = \frac{\text{observed yield}}{\text{theoretical yield}} \times 100$$

$$\text{observed yield} = \frac{30 \times 40}{100 \times 3} = 4 \text{ mole.}$$

Video Solution:

**Q32 Text Solution:**

Moles of CO = Mass/Molar mass
 $= 294/28 = 10.5$ moles
 From the reaction, 3 moles CO forms = 2 moles of Fe
 10.5 moles of CO forms $= 2 \times 10.5/3 = 7$ moles

Video Solution:**Q33 Text Solution:**

Given: 0.220 g CO₂, 0.126 g H₂O, %C = 24

C from CO₂ $= 0.220 \times \frac{12}{44} = 0.060$ g

H from H₂O $= 0.126 \times \frac{2}{18} = 0.014$ g

If 0.060 g C = 24%, total mass =

$0.060 \times 100/24 = 0.25$ g

$$\%H = \frac{0.014}{0.25} \times 100 = 5.6\% = 56 \times 10^{-1}$$

%

Video Solution:**Q34 Text Solution:**

Fluorine reacts to form NaF, which further reacts with CaO (lime).

From the reaction, 1 mole of F₂ needs 1 mole of CaO. Moles of F₂ $= (1900 \times 1000) / 38 = 50000$

mol.

Hence, moles of CaO required = 50000.

Mass of CaO $= 50000 \times 56 = 2800000$ g = 2800 kg.

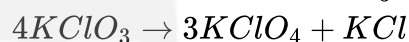
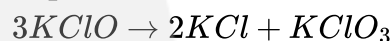
Given lime used = 10000 kg.

So, % utilisation of lime $= (2800 / 10000) \times 100 = 28\%$.

Thus, only 28% of lime was actually used in the reaction based on stoichiometry; rest remained unused. Final answer is 28%.

Video Solution:**Q35 Text Solution:**

First balance the reactions,



By Stoichiometric coefficients, we can calculate the number of moles of potassium chloride and number of moles of potassium chloride will be $3 + 1 + 0.1 = 4.1 \approx 4$.

Video Solution:**Q36 Text Solution:**

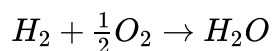
Molecular weight of aspartame

$$= \frac{100}{9.52} \times 14 \times 2 = 294$$

Video Solution:

**Q37 Text Solution:**

$3\text{PbCl}_2 + 2(\text{NH}_4)_3\text{PO}_4 \rightarrow \text{Pb}_3(\text{PO}_4)_2 + 6\text{NH}_4\text{Cl}$
 According to stoichiometry of reaction, 3 mmol of PbCl_2 reacts with 2 mmol of $(\text{NH}_4)_3\text{PO}_4$ to form 1 mmol of $\text{Pb}_3(\text{PO}_4)_2$ and 6 mmol of NH_4Cl .
 So, 72 mmol of PbCl_2 reacts with 48 mmol of $(\text{NH}_4)_3\text{PO}_4$. $\text{PbCl}_2 = \text{L.R.}$
 72 mmol of PbCl_2 will form 24 mmol of $\text{Pb}_3(\text{PO}_4)_2$.

Video Solution:**Q38 Text Solution:**

$$n_{\text{H}_2} = \frac{\omega_{\text{H}_2}}{M_{\text{H}_2}} = \frac{10}{2} = 5 \text{ mole}$$

$$n_{\text{O}_2} = \frac{\omega_{\text{O}_2}}{M_{\text{O}_2}} = \frac{64}{32} = 2 \text{ mole}$$

According to stoichiometry of reaction $\frac{1}{2}$ mole of O_2 require 1 mole of H_2 out form 1 mole of H_2O .
 So, 2 mole of O_2 will require 4 mole of H_2 and form 4 mole of H_2O .

So, amount of water = $4 \times 18 = 72 \text{ g}$

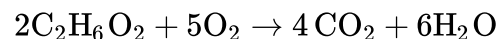
Video Solution:**Q39 Text Solution:**

Let the masses of C, H and O in organic compound X be x, y and z respectively

	Mass			Moles
C	x	4	12	1
H	y	1	3	3
O	z	16/3	16	1

$\therefore$ Empirical formula: CH_3O

Molecular formula: $\text{C}_2\text{H}_6\text{O}_2$



$\therefore$ Number of moles of O_2 required to oxidise 2 moles of (X) = 5.

Video Solution:**Q40 Text Solution:**

3.01×10^{20} molecules of SF_n has mass = $54 \times 10^{-3} \text{ g}$

$$1 \text{ molecules of } \text{SF}_n \text{ has mass} = \frac{54 \times 10^{-3}}{3.01 \times 10^{20}} \text{ g}$$

$$6.02 \times 10^{23} \text{ molecules of } \text{SF}_n \text{ has mass} \\ \frac{54 \times 10^{-3} \times 6.02 \times 10^{23}}{3.01 \times 10^{20}}$$

$$= 108 \text{ g}$$

$$\text{SF}_n = 32 + 19 \times n = 108$$

$$19n = 76$$

$$n = 4$$

Video Solution:**Q41 Text Solution:**

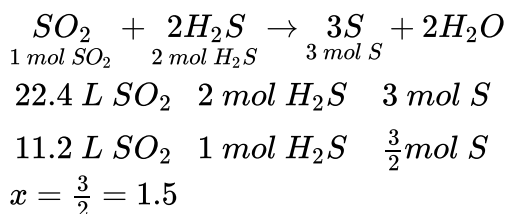
Given: Molar mass of $A_2B = AB_3$

$$\therefore (2A + B)$$

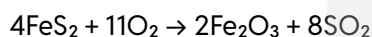
$$= (A + 3B) \left[\begin{array}{l} A \rightarrow \text{Atomic wt. of A} \\ B \rightarrow \text{Atomic wt. of B} \end{array} \right]$$

$$\Rightarrow A = 2B$$

Video Solution:

**Q42 Text Solution:****Video Solution:****Q43 Text Solution:**

Given reaction:



Step 1: O_2 required

For 4 mol $FeS_2 \rightarrow 11$ mol O_2 required

Assume 4 mol FeS_2 taken

Air has 20% O_2 and 80% N_2

Required air = 11 mol $O_2 \rightarrow$ from 55 mol air (since 20% O_2 in air)

50% excess $\rightarrow$ Total air = 55 + 27.5 = 82.5 mol

So, N_2 = 80% of 82.5 = 66 mol

Step 2: Final gas mixture

Gaseous products = SO_2 + unused O_2 + N_2

SO_2 = 8 mol

O_2 used = 11 mol, O_2 in air = $0.2 \times 82.5 = 16.5$ mol

Unused O_2 = 16.5 – 11 = 5.5 mol

N_2 = 66 mol

Total = 8 + 5.5 + 66 = 79.5 mol

% of N_2 = $(66 / 79.5) \times 100 = 83\%$

Answer: 83%

Video Solution:**Q44 Text Solution:**

80% w/w means 80g of solute present in 100 g of solution.

$$n_{NaOH} = \frac{W_{NaOH}}{M_{NaOH}} = \frac{80}{40} = 2 \text{ mole}$$

$$\text{Density} = 2 \text{ g L}^{-1}$$

Mass of solution = 100g

Mass of solvent 100 – 80 = 20 g

$$\begin{aligned}
 \text{Molality} &= \frac{n_{NaOH}}{W_{\text{solvent}}} \times 1000 \\
 &= \frac{2}{20} \times 1000 = 100 \text{ mol kg}^{-1}
 \end{aligned}$$

Video Solution:**Q45 Text Solution:**

100 g alkaloid contains nitrogen = 17.28 g

162 g alkaloid will contain nitrogen

$$= \frac{17.28 \times 162}{100} \text{ g} = 27.9 \text{ g} \approx 28 \text{ g}$$

Atomic weight of nitrogen = 14

So, number of atoms of nitrogen present in one molecule of alkaloid = $\frac{28}{14} = 2$

Video Solution:**Q46 Text Solution:**

Given:

- Volume of solution = 2 L
- Concentration of Mg = 48 ppm = 48 mg/L
- Atomic mass of Mg = 24 g/mol



$$\bullet N_A = 6.02 \times 10^{23}$$

$$\text{Mass of Mg} = 48 \text{ mg/L} \times 2 \text{ L} = 96 \text{ mg} \\ = 0.096 \text{ g}$$

$$\text{Moles of Mg} = \frac{0.096}{24} = 0.004 \text{ mol}$$

$$\text{Number of atoms} = 0.004 \times 6.02 \times 10^{23} \\ = 2.408 \times 10^{21}$$

$$2.408 \times 10^{21} = 24.08 \times 10^{20}$$

$$x = 24$$

Video Solution:



Q47 Text Solution:

Let's say 100 gm of herbal paste is given initially
25 gm water

$\therefore$ Dry mass (including active comp.) = 75 g

After drying $\text{H}_2\text{O} = 5\%$

$\therefore$ New partially dried sample = x g

This x g partial dry sample contain 75 g of dry mass.

$$0.95x = 75$$

$$x = \frac{75}{0.95} = 78.95$$

Active compound is 70 % of dried mass

$$\therefore \text{Mass} = \frac{70}{100} \times 78.95 = 55.265 \text{ g}$$

$\therefore$ % of active compound in original sample

$$= \frac{55.26}{100} \times 100 = 55.265 \%$$

Video Solution:



Q48 Text Solution:

Let percent abundance of A_3 be x

Percent abundance of $A_2 = 3x$

Percent abundance of $A_1 = 100 - x - 3x$

$$1.25 = \frac{(1 \times (100 - 4x)) + (3x \times 2) + (x \times 3)}{100}$$

On solving, $x = 5$

$$A_1 = 100 - 4x = 100 - 20 = 80 \%$$

Video Solution:



Q49 Text Solution:



$$233 \text{ g of BaSO}_4 = 174 \text{ g of K}_2\text{SO}_4$$

$$1.2 \text{ g of BaSO}_4 = \frac{174}{233} \times 1.2 = 0.8961 \text{ g}$$

5g of K_2SO_4 are present in 250 mL of sol.

$$0.8961 \text{ g of K}_2\text{SO}_4 \text{ are present in } \frac{250}{5} \times 0.8961$$

$$V = 44.8 \text{ mL} \simeq 45 \text{ mL}$$

Video Solution:



Q50 Text Solution:

Consider, $V = 1\text{L} \Rightarrow n_B = 7$ moles

Mass of solution = 1140 gm.

$$\Rightarrow M_A + n_{\text{H}_2\text{O}} = 1140 - 7 \times 30 = 930 \text{ g}$$

$$10n_A = 18n_{\text{H}_2\text{O}} = 930$$

$$\text{Also, } X_A = \frac{n_A}{n_A + 7 + n_{\text{H}_2\text{O}}} = 0.05$$

Solving, $n_A = 3$ mole

Video Solution:





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